1 Asymmetric KV budget

Split the index-storage budget

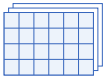
$$B_{KV} = B_K + B_V$$

K: 6 bits (avg.) | V: 2

8 index bits per K/V pair on average

Keys (K)

Attention weights



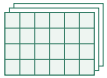
$T \times d_{head}$

K4 / K8

6 bits on average

Values (V)

Weighted sum



$T \times d_{head}$

V2

2 bits per value

One matrix per KV head

2 Boundary allocation

28-layer Qwen3-1.7B

Layers 1–7
Self-attention | FFN
Values: V2 — K8

Layers 8–21
Self-attention | FFN
Values: V2 — K4

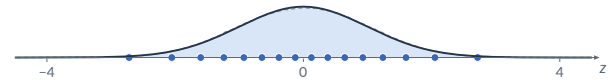
Layers 22–28
Self-attention | FFN
Values: V2 — K8

K8: 14 layers · K4: 14 layers
V2 in every layer

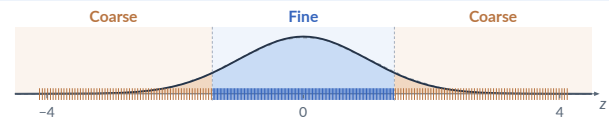
3 Hardware-aware KV codecs

Gaussian model | KV samples | Reconstruction levels

K4 · Lloyd–Max — 16 levels



K8 · PU-K8 (ours) — 256 levels



Two spacings: $\Delta_0 < \Delta_1$

V2 · Lloyd–Max — 4 levels

